

M1 Gate Results: Reproducing Nanda et al.'s Grokking

Task: $(a + b) \bmod 97$ | Jonathan John | Supervisor: Prof. Dr.-Ing. Sheikh Faisal Rashid
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Objective

Reproduce the canonical grokking phenomenon reported by Nanda et al. on the modular addition task $(a + b) \bmod 97$, as the mandatory M1 gate before benchmarking the nine grokking predictors (L2 Norm, Dropout, Spectral, AGE, HTSR Alpha, Correlation Traps, Weight-PCA, Higher-MI, Commutator Defect).

Experimental Setup

Task	Modular addition, $(a + b) \bmod 97$; target predicted at the "=" token position
Architecture	Transformer following Nanda et al.'s simplified architecture (vocab_size=98, d_model=128)
Optimizer	AdamW (lr = 1e-3, weight_decay = 1.0)
Loss	Cross-entropy on the logit at the "=" position
Data split	30% train / 70% held-out test, full-batch training
Training length	20,000 epochs
Hardware	Apple Silicon (MPS), end-to-end device pipeline verified

Results

Train accuracy reaches 100% by roughly epoch 800 (pure memorization phase). Test accuracy remains near chance level (~0%) for over 2,000 further epochs — the characteristic generalization lag — before climbing sharply from approximately epoch 3,000 to epoch 5,000, reaching near-100% test accuracy. Both curves remain flat at their converged values through the remainder of training (20,000 epochs total).

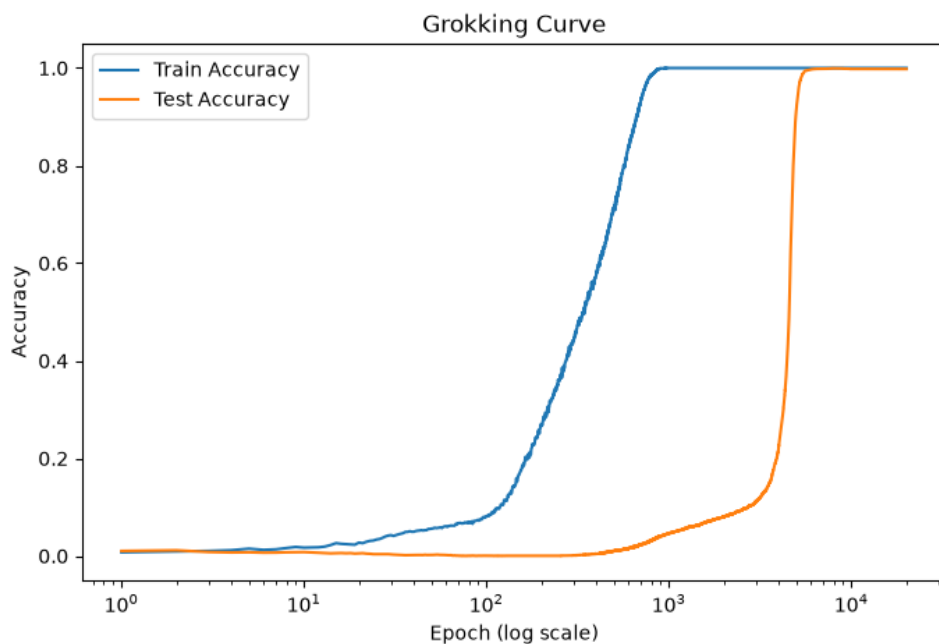


Figure 1. Train vs. test accuracy over training, log-scale epoch axis.

Metric	Final Value
Train Accuracy	1.000 (100%)
Test Accuracy	0.998 (99.8%)

Interpretation

This matches Nanda et al.'s canonical grokking signature: near-perfect memorization of the training set long before generalization, followed by a delayed and abrupt jump in test accuracy. The M1 gate is confirmed closed, unblocking predictor benchmarking work.

Code

The training pipeline (data generation, transformer model, training loop, and plotting) is provided separately alongside this report.

Next Steps

Proceed to predictor benchmarking, beginning with the L2 Norm predictor (first in the nine-predictor evaluation order).